

TONGCHUAN, China

WMO: 539470

Lat: 35.083N

Lon: 109.067E

Elev: 914

StdP: 90.82

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-8.7	-23.6	0.5	-3.3	-21.0	0.6	-2.6	7.4	1.3	6.4	1.7	1.6	40	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	31.7	19.4	30.4	19.3	28.9	19.4	23.2	27.5	22.4	26.6	21.7	25.7	3.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	18.7	25.2	21.3	17.8	24.8	20.4	16.9	24.2	74.4	27.7	71.4	26.4	68.5	25.8	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.0	5.8	5.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-2.1	0.8	5.4	11.7	17.5	21.3	23.0	22.1	16.5	11.6	5.2	-0.7
	DBStd	9.35	3.01	3.26	3.75	3.75	3.26	2.86	2.46	2.58	2.69	3.10	3.97	3.24
	HDD10.0	1303	376	258	148	24	1	0	0	0	0	17	148	332
	HDD18.3	3051	634	491	401	199	56	7	1	2	66	211	393	590
	CDD10.0	1697	0	0	6	76	232	338	404	376	196	65	4	0
	CDD18.3	404	0	0	0	1	28	96	147	120	11	1	0	0
	CDH23.3	3035	0	0	0	24	267	877	1071	731	61	6	0	0
	CDH26.7	901	0	0	0	3	53	297	354	188	6	0	0	0
Wind	WSAvg	2.2	2.1	2.2	2.6	2.6	2.5	2.3	2.1	2.2	1.9	2.1	2.2	2.1
Precipitation	PrecAvg	595	4	8	23	42	53	56	115	120	98	51	21	5
	PrecMax	889	26	27	62	87	135	139	210	345	232	151	72	19
	PrecMin	354	0	0	0	5	3	3	42	19	21	7	0	0
	PrecStd	116	5	7	14	22	37	32	47	76	62	37	16	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	12.9	19.2	26.5	30.5	33.2	33.5	33.3	28.1	25.8	17.7	10.1
		MCWB	1.4	4.6	10.3	14.7	16.7	17.8	19.6	20.0	17.8	15.0	9.4	2.7
	2%	DB	6.7	10.1	16.5	23.2	28.1	31.5	31.6	30.6	26.1	21.6	15.2	7.9
		MCWB	0.2	3.0	8.5	13.4	15.7	17.5	20.5	20.7	17.8	14.4	8.7	1.7
	5%	DB	4.8	8.1	14.3	21.0	26.3	30.0	30.4	28.9	23.9	19.5	13.0	6.0
		MCWB	-0.7	2.0	7.2	12.1	15.6	17.4	20.5	21.1	17.2	13.3	7.5	0.8
	10%	DB	3.1	6.4	12.1	18.7	24.5	28.2	28.9	27.5	22.0	17.1	11.0	4.1
		MCWB	-1.7	1.0	5.8	11.0	15.2	17.2	20.4	20.9	16.6	11.9	6.9	-0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	5.4	11.3	16.7	19.6	21.8	24.4	24.0	21.0	17.1	11.1	4.1
		MCDB	8.0	11.1	17.1	23.7	25.6	28.8	28.9	28.3	26.6	22.7	14.9	8.2
	2%	WB	0.7	3.8	9.5	14.9	18.6	20.3	23.4	23.2	19.5	15.2	9.8	2.4
		MCDB	5.9	8.3	15.0	21.4	24.4	26.3	27.7	27.6	23.2	19.8	13.2	6.5
	5%	WB	-0.4	2.7	8.1	13.3	17.7	19.5	22.6	22.5	18.4	14.0	8.6	1.3
		MCDB	3.8	6.8	12.7	19.0	23.1	25.5	26.8	26.6	21.8	18.0	11.6	4.9
	10%	WB	-1.3	1.7	6.8	12.0	16.7	18.7	21.9	21.9	17.5	13.0	7.6	0.1
		MCDB	2.2	5.7	11.3	17.4	21.6	24.4	26.1	25.5	20.7	16.2	10.2	3.5
Mean Daily Temperature Range	5% DB	MDBR	8.6	8.8	9.5	9.9	11.1	10.8	8.6	8.1	7.8	8.6	8.4	8.6
		MCDBR	11.1	11.7	13.6	13.5	14.6	14.6	12.1	10.7	12.0	11.4	11.7	11.5
		MCWBR	7.1	6.9	7.7	6.7	6.6	5.3	4.2	4.3	5.8	6.0	7.2	7.0
	5% WB	MCDBR	10.4	9.6	11.8	12.7	11.4	11.0	9.1	7.9	9.3	9.6	10.0	10.6
		MCWBR	6.8	5.9	7.3	7.5	6.2	4.9	4.6	4.2	5.6	5.8	6.5	6.7
Clear-Sky Solar Irradiance	taub	0.494	0.584	0.605	0.610	0.597	0.635	0.656	0.688	0.630	0.602	0.525	0.460	
	taud	1.748	1.600	1.599	1.628	1.711	1.677	1.685	1.605	1.676	1.666	1.746	1.833	
	Ebn at Noon	650	636	676	705	723	694	676	641	650	614	615	654	
	Edh at Noon	177	231	250	254	237	245	241	257	228	213	177	154	
All-Sky Solar Radiation	RadAvg	2.60	3.11	4.18	4.99	5.39	5.53	5.15	4.50	3.57	3.13	2.67	2.42	
	RadStd	0.26	0.48	0.42	0.45	0.41	0.48	0.63	0.57	0.39	0.48	0.38	0.29	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-73	-68

Nomenclature: See separate page